

Timeline Proposal

Date Range	Objectives and Deliverables
Phase I: Development	
Mar 02 – Mar 15	Architecture & Data Pipeline <i>Backend:</i> Implementation of MNE pipeline to parse .gdf files and export raw JSON. <i>Frontend:</i> Setup of Three.js scene; implementation of JSON loader.
Mar 16 – Mar 29	3D Asset Integration <i>Backend:</i> Application of 8–30 Hz bandpass filters (Mu/Beta rhythms). <i>Frontend:</i> Import of skull-stripped brain mesh; mapping of 22 electrodes to 3D surface coordinates.
Mar 30 – Apr 12	Signal Animation <i>Integration:</i> Connection of pre-processed signal arrays to the rendering loop. <i>Frontend:</i> Implementation of visual feedback (color/intensity changes) driven by signal amplitude.
Mar 13 – Apr 26	UI Dashboard & Code Freeze <i>UI/UX:</i> Implementation of „Time-Travel“ scrubber. <i>Milestone:</i> Code Freeze upon completion of the prototype.
Phase II: Documentation & Evaluation	
Apr 27 – May 10	Methodology & Architecture Drafting of Section 3 (Methodology) and Section 1 (Introduction). Creation of high-resolution figures from the completed prototype.
May 11 – May 24	Implementation & Evaluation Drafting of Section 4 (Implementation Steps) and qualitative evaluation. Documentation of performance metrics (FPS) and data processing steps.
May 25 – Jun 07	Full Draft Assembly Completion of Abstract, Conclusion, and Bibliography. <i>Milestone:</i> Submission of the first full draft to the supervisor for feedback.
Jun 08 – Jun 30	Review & Submission Incorporation of supervisor feedback. Final formatting checks and proofreading. <i>Milestone:</i> Final Thesis Submission.

Table 1: Proposed Project Timeline